

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination
2014

CL211 Environmental Engineering-I

Date: 9th May, 2014

Day: Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Write short note on: "Water Borne Diseases". [03]
(b) Explain the role of environmental engineer. [04]
- Q - 2 (a) List the physical characteristics of water. Explain any one in detail. [03]
(b) Draw the layout of surface water treatment plant. Mention the function of each unit. [04]
(c) Explain theory of filtration. [03]
(d) What is coagulation? Explain the theory of coagulation. [04]

OR

- Q - 2 (a) Differentiate between Slow Sand filter and Rapid Sand Filter. [03]
(b) Explain the backwashing of rapid sand filter. [04]
(c) Explain "Super Chlorination". [03]
(d) Compare the characteristics of different coagulants used in water treatment. [04]
- Q - 3 Attempt any Two: [14]
(a) Design a circular sedimentation tank filled with mechanical sludge remover for water works which has to supply daily 3.5 million liters of water to the town. The detention period in the tank for water for 4 hours and depth of water in the tank may be assumed as 3 m.
(b) Give classification of hardness. Describe the method for removal of hardness by lime-soda process.
(c) Describe in details the working of Pressure filter.

SECTION – II

- Q - 4 (a) What are the requirements of wholesome water? [03]
(b) Explain the various factors affecting the per capita demand. [04]
- Q - 5 (a) Draw a neat sketch of any one type of intake structure. [03]
(b) Draw the neat sketch showing the house service connection from the distribution main and state the function of any three components. [04]
(c) Explain intermittent system of water supply. [03]
(d) Classify the traps used in house drainage. [04]

OR

- Q - 5 (a) What are points should be kept in mind while selecting a site for intake structure? [03]
(b) Draw the neat sketch of overhead tank and state the function of each component. [04]
(c) What are the requirements of a distribution system? [03]
(d) Enlist systems of ventilation. Explain any one. [04]
- Q - 6 **Attempt any Two.** [14]
(a) Explain the methods of refuse disposal.
(b) Explain different sanitary fittings in house drainage system.
(c) Give importance of air conditioning and lighting in a ventilation system.
